

# Problem Set 3

## Applied Stats II

Due: March 25, 2026

### Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in .pdf form.
- This problem set is due before 23:59 on Wednesday March 25, 2026. No late assignments will be accepted.

### Question 1

We are interested in how governments' management of public resources impacts economic prosperity. Our data come from Alvarez, Cheibub, Limongi, and Przeworski (1996) and is labelled `gdpChange.csv` on GitHub. The dataset covers 135 countries observed between 1950 or the year of independence or the first year for which data on economic growth are available ("entry year"), and 1990 or the last year for which data on economic growth are available ("exit year"). The unit of analysis is a particular country during a particular year, for a total  $> 3,500$  observations.

- Response variable:
  - `GDPWdiff`: Difference in GDP between year  $t$  and  $t-1$ . Possible categories include: "positive", "negative", or "no change"
- Explanatory variables:
  - `REG`: 1=Democracy; 0=Non-Democracy
  - `OIL`: 1=if the average ratio of fuel exports to total exports in 1984-86 exceeded 50%; 0= otherwise

Please answer the following questions:

1. Construct and interpret an unordered multinomial logit with `GDPWdiff` as the output and "no change" as the reference category, including the estimated cutoff points and coefficients.

```
1 #####
2 # Problem 1
3 #####
4
5 # load data
6 gdp_data <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsII_
   2026/main/datasets/gdpChange.csv", stringsAsFactors = F)
7
8 #dropping missing values
9 gdp_q1 <- na.omit(gdp_data[, c("GDPWdiff", "REG", "OIL")])
10
11 # creating 3 outcome categories for change in GDP
12 gdp_q1$GDPWdiff_cat <- ifelse(gdp_q1$GDPWdiff < 0, "negative",
   ifelse(gdp_q1$GDPWdiff > 0, "positive", "no
   change"))
14
15 table(gdp_q1$GDPWdiff_cat)
16
17
18
19 # 1) Unordered multinomial logit Reference category: "no change"
20 gdp_q1$GDPWdiff_unordered <- relevel(
21   factor(trimws(gdp_q1$GDPWdiff_cat),
22     levels = c("no change", "negative", "positive")),
23   ref = "no change"
24 )
25
26 mnl_mod <- multinom(GDPWdiff_unordered ~ REG + OIL,
27   data = gdp_q1,
28   trace = FALSE)
29 summary(mnl_mod)
30
31 # coefficient table with z-statistics and p-values
32 mnl_sum <- summary(mnl_mod)
33 mnl_coef <- mnl_sum$coefficients
34 mnl_se <- mnl_sum$standard.errors
35 mnl_z <- mnl_coef / mnl_se
36 mnl_p <- 2 * (1 - pnorm(abs(mnl_z)))
37
38 mnl_results <- round(cbind(
39   Estimate = c(mnl_coef),
40   Std.Error = c(mnl_se),
41   z.value = c(mnl_z),
```

```

42 p.value = c(mnl_p)
43 ), 4)
44
45 mnl_results
46
47 # category-specific intercepts and slopes
48 round(mnl_coef, 4)
49
50 # relative risk odds ratios for category-specific slopes
51 exp(round(mnl_coef, 4))

```

Table 1: Unordered Multinomial Logit Model

|                   | <i>Dependent variable:</i>                           |                                   |
|-------------------|------------------------------------------------------|-----------------------------------|
|                   | GDPWdiff (ref. = no change)<br>negative vs no change | positive<br>positive vs no change |
|                   | (1)                                                  | (2)                               |
| REG               | 1.379*<br>(0.769)                                    | 1.769**<br>(0.767)                |
| OIL               | 4.784<br>(6.885)                                     | 4.576<br>(6.885)                  |
| Constant          | 3.805***<br>(0.271)                                  | 4.534***<br>(0.269)               |
| Akaike Inf. Crit. | 4,690.770                                            | 4,690.770                         |

*Note:*

\*p<0.1; \*\*p<0.05; \*\*\*p<0.01

Democracy (REG) is positively associated with both positive and negative GDP changes relative to no change, but the effect is only statistically significant for positive outcomes. Oil dependence (OIL) is not statistically significant in either comparison, and the large standard errors suggest considerable uncertainty. Because the reference category (no change) is very small, these results should be treated with caution.

2. Construct and interpret an ordered multinomial logit with GDPWdiff as the outcome variable, including the estimated cutoff points and coefficients.

```

1 # 2) Ordered multinomial logit Ordering
2
3
4 gdp_q1$GDPWdiff_ordered <- ordered(
5   trimws(gdp_q1$GDPWdiff_cat),
6   levels = c("negative", "no change", "positive")
7 )
8
9 ologit_mod <- polr(GDPWdiff_ordered ~ REG + OIL,
10                   data = gdp_q1,
11                   method = "logistic",
12                   Hess = TRUE)
13
14 summary(ologit_mod)
15
16 # coefficient table with p-values
17 ologit_ct <- coef(summary(ologit_mod))
18 ologit_p <- 2 * (1 - pnorm(abs(ologit_ct[, "t value"])))
19
20 ologit_results <- round(cbind(
21   ologit_ct,
22   p.value = ologit_p
23 ), 4)
24
25 ologit_results
26
27 # ordered-logit cut points
28 round(ologit_mod$zeta, 4)
29
30 stargazer(
31   mnl_mod,
32   type = "latex",
33   title = "Unordered Multinomial Logit Model",
34   dep.var.labels = "GDPWdiff (ref. = no change)",
35   column.labels = c("negative vs no change", "positive vs no change"),
36   digits = 3,
37   single.row = FALSE
38 )
39
40 stargazer(
41   ologit_mod,
42   type = "latex",
43   title = "Ordered Multinomial Logit Model",
44   dep.var.labels = "GDPWdiff (negative < no change < positive)",
45   digits = 3,
46   single.row = FALSE
47 )

```

The ordered model provides a clearer pattern. Democracy has a strong positive effect, indicating that democratic countries are more likely to experience better economic outcomes overall. In contrast, oil dependence is

Table 2: Ordered Multinomial Logit Model

|                                          | <i>Dependent variable:</i>                 |
|------------------------------------------|--------------------------------------------|
|                                          | GDPWdiff (negative < no change < positive) |
| REG                                      | 0.398***<br>(0.075)                        |
| OIL                                      | -0.199*<br>(0.116)                         |
| Observations                             | 3,721                                      |
| <i>Note:</i> *p<0.1; **p<0.05; ***p<0.01 |                                            |

associated with worse outcomes, although this effect is only weakly significant. This suggests some evidence that reliance on oil may hinder growth, but the result is not very robust.

Overall, the ordered model is more reliable and shows that democracy is linked to improved economic performance, while the evidence for oil dependence is weaker and less certain.

## Question 2

Consider the data set `MexicoMuniData.csv`, which includes municipal-level information from Mexico. The outcome of interest is the number of times the winning PAN presidential candidate in 2006 (`PAN.visits.06`) visited a district leading up to the 2009 federal elections, which is a count. Our main predictor of interest is whether the district was highly contested, or whether it was not (the PAN or their opponents have electoral security) in the previous federal elections during 2000 (`competitive.district`), which is binary (1=close/swing district, 0="safe seat"). We also include `marginality.06` (a measure of poverty) and `PAN.governor.06` (a dummy for whether the state has a PAN-affiliated governor) as additional control variables.

- (a) Run a Poisson regression because the outcome is a count variable. Is there evidence that PAN presidential candidates visit swing districts more? Provide a test statistic and p-value.

```

1 #####
2 # Problem 2
3 #####
4
5 # load data

```

```

6 mexico_elections <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/
  StatsII_2026/main/datasets/MexicoMuniData.csv")
7
8 # keep only variables needed for Q2 and drop missing values
9 mexico_q2 <- na.omit(mexico_elections[, c(
10   "PAN.visits.06",
11   "competitive.district",
12   "marginality.06",
13   "PAN.governor.06"
14 )])
15
16 summary(mexico_q2$PAN.visits.06)
17 table(mexico_q2$competitive.district)
18 table(mexico_q2$PAN.governor.06)
19 summary(mexico_q2$marginality.06)
20
21 # Poisson regression
22
23 pois_mod <- glm(
24   PAN.visits.06 ~ competitive.district + marginality.06 + PAN.governor
    .06,
25   data = mexico_q2,
26   family = poisson(link = "log")
27 )
28
29 summary(pois_mod)
30
31 # coefficient table
32 pois_ct <- coef(summary(pois_mod))
33 pois_results <- round(cbind(
34   pois_ct,
35   IRR = exp(coef(pois_mod))
36 ), 4)
37
38 pois_results
39
40
41 # (a): test for competitive districts
42
43 # Extract z-statistic and p-value for competitive.district
44 comp_z <- coef(summary(pois_mod))["competitive.district", "z value"]
45 comp_p <- coef(summary(pois_mod))["competitive.district", "Pr(>|z|)"]
46
47 comp_z
48 comp_p

```

The coefficient on competitive.district is  $-0.081$  ( $z = -0.477$ ,  $p = 0.634$ ), providing no evidence that PAN candidates visited competitive districts more often. The incidence-rate ratio is  $e^{-0.081} \approx 0.92$ , suggesting about an 8% lower expected number of visits in competitive districts, though this

effect is not statistically significant.

- (b) Interpret the `marginality.06` and `PAN.governor.06` coefficients.

```
1 # (b): exponentiated coefficients
2
3 exp(coef(pois_mod))
```

The coefficient on `marginality.06` is  $-2.080$  ( $p < 0.001$ ), indicating that higher `marginality` is associated with fewer PAN candidate visits. The incidence-rate ratio is  $e^{-2.080} \approx 0.125$ , implying about an 87.5% decrease in expected visits for a one-unit increase. The coefficient on `PAN.governor.06` is  $-0.312$  ( $p = 0.062$ ), suggesting that municipalities in states with a PAN governor received fewer visits. The incidence-rate ratio is  $e^{-0.312} \approx 0.73$ , indicating roughly a 27% reduction in expected visits, though this effect is only marginally significant at the 10% level.

The coefficient on `PAN.governor.06` is  $-0.312$  ( $p = 0.062$ ), indicating fewer expected visits in municipalities with a PAN governor. The incidence-rate ratio is  $e^{-0.312} \approx 0.73$ , corresponding to about a 27% decrease, though the effect is only marginally significant at the 10% level.

- (c) Provide the estimated mean number of visits from the winning PAN presidential candidate for a hypothetical district that was competitive (`competitive.district=1`), had an average poverty level (`marginality.06 = 0`), and a PAN governor (`PAN.governor.06=1`).

```
1 #(c): predicted mean number of visits
2 new_case <- data.frame(
3   competitive.district = 1,
4   marginality.06 = 0,
5   PAN.governor.06 = 1
6 )
7 # predicted expected count
8 predict(pois_mod, newdata = new_case, type = "response")
9
10
11 stargazer(
12   pois_mod,
13   type = "latex",
14   title = "Poisson Regression Model",
15   dep.var.labels = "PAN candidate visits in 2006",
16   digits = 3,
17   single.row = FALSE
18 )
```

For a district with `competitive.district = 1`, `marginality.06 = 0`, and `PAN.governor.06 = 1`, the model predicts an expected mean of 0.01495 PAN candidate visits (approximately 0.015). This represents an expected count rather than an actual observed number of visits.

Table 3: Poisson Regression Model

|                      | <i>Dependent variable:</i>   |
|----------------------|------------------------------|
|                      | PAN candidate visits in 2006 |
| competitive.district | −0.081<br>(0.171)            |
| marginality.06       | −2.080***<br>(0.117)         |
| PAN.governor.06      | −0.312*<br>(0.167)           |
| Constant             | −3.810***<br>(0.222)         |
| Observations         | 2,407                        |
| Log Likelihood       | −645.606                     |
| Akaike Inf. Crit.    | 1,299.213                    |
| <i>Note:</i>         | *p<0.1; **p<0.05; ***p<0.01  |